What is health economics? Describe the relevance of health economics/why do we study health economics?

Health economics:

Health economics is a branch of economics concerned with issues related to efficiency, effectiveness, value and behaviour in the production and consumption of health and health care. Health economics studies how resources are allocated to and within the health economy. The production of health care and its distribution across population falls within this definition. Kenneth Arrow is the father of health economics.

The relevance of health economics:

The study of health economics is important and interesting in three related ways:

(i) the size of the contribution of health sector to the overall economy;

(ii) the national policy concerns resulting from the importance

Paper Source
Subject.........
Date:.........Time:.........

many people attach to the economic problems they face in pursuing and maintaining their health and

(iii) the many health issues that have a substantial economic element.

Features of economic analysis :

Many distinctive features of economics might be indentified but we emphasize four.

(i) Scarcity of social resources
(ii) Assumption of rational decision making
(iii) Concept of marginal analysis
(iv) Use of economic models.

(i) Scarcity of social resources:

Economic analysis is based on the premise that individuals must give up some of one resource in order to get some of another. At the national level, this means that increasing shares of GDP going to health care ultimately imply decreasing shares available for other uses. Individuals sell their time for wages, and individuals will refuse overtime work even if offered more than their normal wage rate — because "it's not worth it."

(ii) Rational decision making:

Economists typically approach problems of human economic behavior by assuming that the decision maker is a rational being. Rationality is affectively defined as "making choices that best further one's own ends given one's resource constraints." Some behaviors may appear irrational.

(iii) Marginal analysis:

Mainstream economic analyses features reasoning at the margin. To make an appropriate choice, decision makers must understand the cost as well as the benefit of the next, or marginal ~~cost~~ unit.

(iv) Use of models:

Finally, economics characteristically develops models to depict its subject matter. The models may be described in words, graphs or mathematics. In economic analysis, the models are often abstract which help to make sence of the world, in economics an in everyday life.

++ In broad terms, health economics study the functioning ~~for functioning~~ of the health care systems as well as health affecting behaviors such as somking. Health economics is a sub-discipline of economics that is concerned with the efficient ~~dicision~~ allocation of health care resources.

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Paper Source
Subject.........
Date:......... Time:.........

Does economics apply to health and health care? Does price matter?

If economics studies how scarce resources are used to produce goods and services and then how these goods services are distributed, then certainly economics applies to health and health care. Certainly, health care resources are scarce; in fact, their cost concerns most people. There is no question that health care is produced and distributed.

However, much of health care simply does not fit some emergency image. A considerable amount of health care is elective, meaning that patients have and will percive some choice over whether and when to have the diagnostic or treatment involved. Much health care even routine, involving problems such as upper respiratory infections, back pain and diagnostics check ups. The patient often has prior experience with these concerns. Furthermore, even in a real emergency, the consumers have agents to make or help make decisions on their behalf. Traditionally physicians have served as agents and more recently, care managers have also entered the process. Thus, rational choices ~~are~~ can be made.

Does price matter?:

Many have argued that health care is so different from other goods that consumers do not respond to financial incentives. Following figure examines the use of ambulatory mental health and medical care where amounts of health care consumed are are measured along the horizontal axis. These amounts are scaled in percentage terms from zero to 100 percent, where 100 percent reflects the average level of care consumed by the group that used the most care on average.

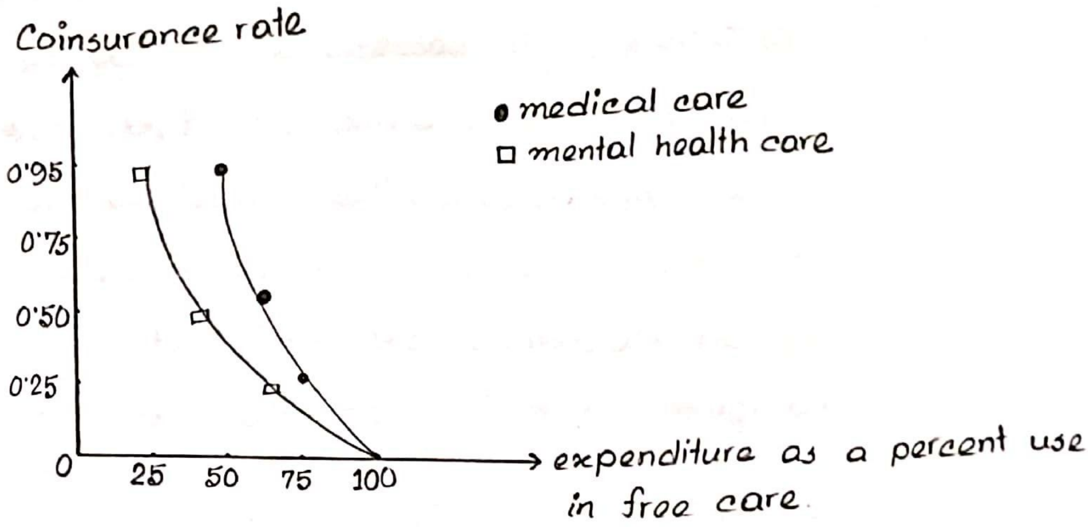


Fig: Demand response of ambulatory mental and medical care

This group is the group with free care. The vertical axis measures the economic ~~inten~~ incentives as indicated by the coinsurance rate – the percentage of the bill paid out

Paper Source
Subject..........
Date:..........Time:..........

directly by the consumer. Thus a higher coinsurance rate reflects a higher price to the consumer. The above figure shows people consuming more care as the care becomes less costly in terms of dollars paid out-of-pocket. More importantly, the curve demonstrates that economic ~~inten~~ incentives do matter. Those facing higher prices demand less care.

Paper Source
Subject.......
Date:.......Time:.......

Is health care different? Why it is different from other goods and services. Explain.

Although economics certainly applies to health care, it is more challenging to answer the question of how directly and simply it applies. Health care is different for the following reasons. —

(i) Presence and Extent of Uncertainty:

When Nobel laureate Kenneth Arrow (1963) directed his attention to the economics of health, he helped establish health economics as a field. He stressed the prevalance of uncertainty in health care, on both the demand side and supply side. Consumers are uncertain of their health status and need for health care in any coming period. This means that the demand for health care is irregular in nature from the individual's perspective; likewise, the demand facing a health care firm is irregular.

Uncertainty is also prevalent on the supply side. Standard economic analysis often assumes the products, and the pleasures that they bring, are well understood by the purchasers. The purchase of steak, milk, new clothes etc provides expected well being that is easily known and unterstood. In contrast, several cases of product uncertainty exist in the health field. Consumers often do not know the expected outcomes of various treatments without physicians' advice, and in many cases physicians themselves cannot predict the outcomes of treatments with certainty.

(ii). Prominence of Insurance:

Consumers purchase insurance to guard against the uncertainty and risk. Because health insurances pay directly for the full cost of people health care.

Suppose, in a specific year almost 52 percent of all personal health care expenditures were paid out of pocket meaning that 48 percent was paid by third party payers. Insurance changes the demand for care and it potentially also changes the incentives of providers.

(iii). Problems of Information:.

Uncertainty can in part be attributed to lack of information. Sometimes information is unavailable to all parties concerned. The information in equestion is known to some persons or parties but not to all, and then it is the asymmetry of information that is problematic.

The problems of information mean that careful economic analysis must modify their methods. They bare make their decisions on the characteristics of the goods, theirs prices and their abilities to bring pleasure to the buyers.

Consumers may not know which physicians or hospitals are good, capable or even compelent. They may not know whether they are ill or what should be done if they are. This lack of information often makes a consumer depend on the provider. The provider offers both the information and the service leading to the possibility of conflicting interests.

(iv) Large Role of Non-profit firm:

Economists often assume that firms maximizes profits. Yet many health care providers, includings many ~~hop~~ hospitals, insurances and nursing homes, have nonprofit status. The economists must analyze the establishment and ~~perpetution~~ perpetuation of nonprofit institutions and understand the differences in their behaviours from non-profit firms. The public and the larger medical community are aware of the major hospitals as centres of health care, teaching and research.

(v) Restrictions on competition:

The health sector has developed many practices that effectively restrict competition. These practices include licensure requirments for providers; restrictions on provider advertising, and standards of ethical behaviour that enjoin providers from competing with each other. Regulation to promote quality or curb costs also reduce the freedom of choice of providers and may influence competition.

(vi) Role of Equity and Need:

Poor health of another human being often envokes a feeling of concern that distinguishes health care from many other goods and services. Many advocates express this feeling by saying that people ought to get health care they need regardless of whether they can afford it.

(vii) Government Subsidies and Public Provision:

In most countries, the government plays a major role in the provision of financing of health services.

Using supply and demand graph and assuming competitive markets, show and explain the effect on equilibrium price and quantity of the following:-

(a) A decrease of the price of Ace on the market for Napa. (Hint: Ace and Napa are substitutes)

Since Ace and Napa are substitute goods. A decrease of the price of Ace will decrease the demand for Napa.

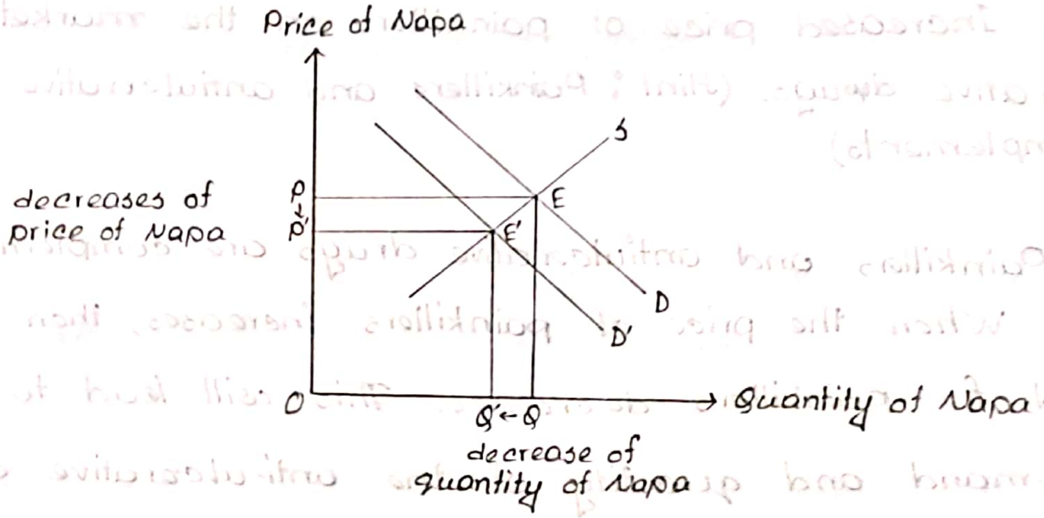


Fig:(1) Effect of decreased price Ace and Market for Napa.

As the price of Ace decreases, the demand for Ace increases. On the other hand, the demand for Napa decreases.

because Napa and Ace are substitutes. Initially, the equilibrium was point E, where demand for Napa was Q and price was P.

As demand for Napa decreases, the demand curve shifts to the left from D to D'. Therefore, the quantity of Napa decreases from Q to Q'. The equilibrium position of the competitive market shifted from E to E' and price decreases from P to P'. These are shown in fig 1.

(b) Increased price of painkillers on the market for antiulcerative drugs. (Hint : Painkillers and antiulcerative drugs are complements).

Painkillers and antiulcerative drugs are complementory goods. When the price of painkillers increases, then the demand for painkillers decreases. This will lead to decrease the demand and quantity of the anti-ulcerative drugs in the market.

Therefore, the demand curve of the anti-ulcerative drugs shifts left from D to D'. The quantity

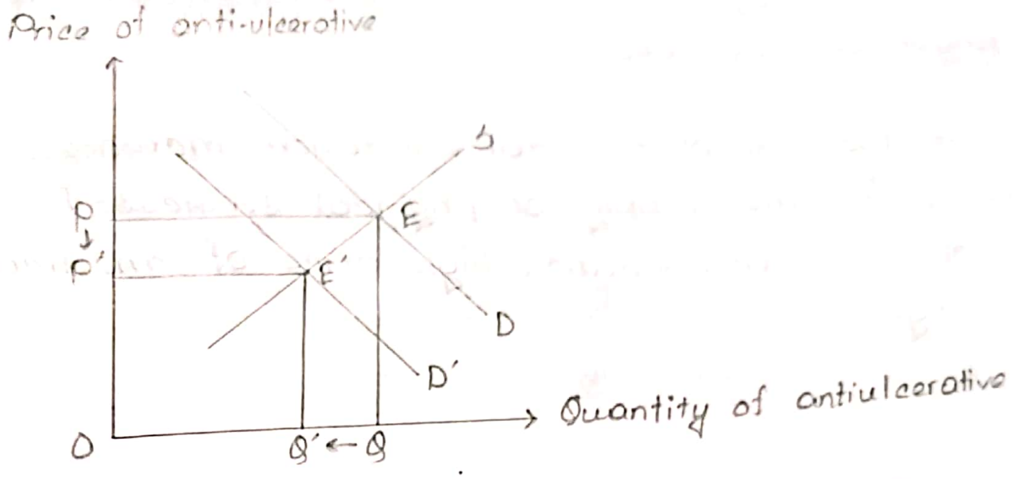


Fig: (2) Effect of increased price of painkillers and market for anti-ulcerative drugs

of anti-ulcerative decreases from Q to Q'. The equilibrium shifts from E to E' and price decreases from P to P'. These are shown in fig (2).

Paper Source
Subject.
Date: Time:

(c) Increased cost of medical education on the market for physician services.

If the cost of medical education increases, it will lead to decrease in the supply of physical services. Because of high cost of medical education, high price of medical services is charged.

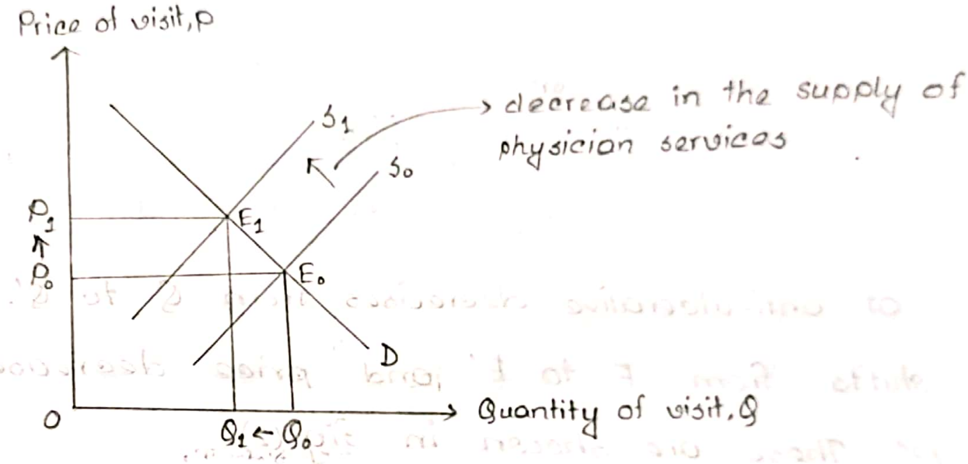


Fig (3): Market for physician services

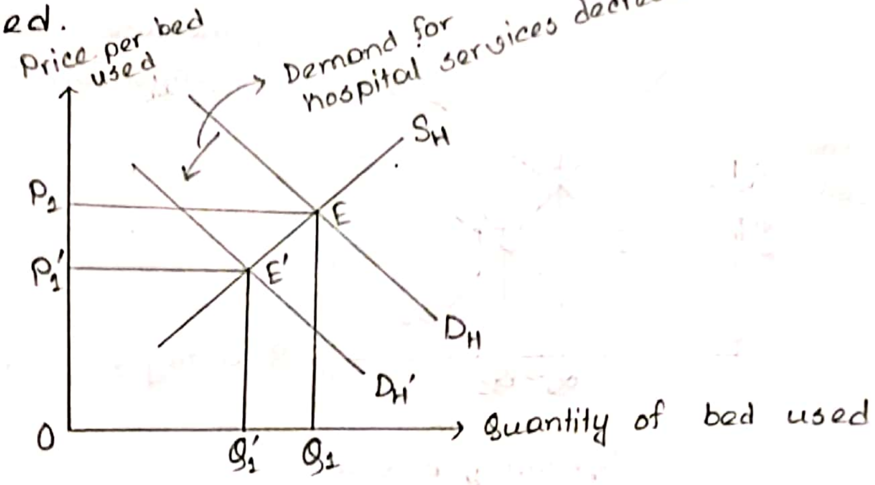
When the cost of medical education increases, the supply of physician service or medical service decreases. This shifts the supply curve to the left from S_0 to S_1. The initial equilibrium point $E(Q_0, P_0)$, where the quantity of visit was Q_0 and price was P_0.

After the decrease of the supply of physician services, there creates a new equilibrium E' where the quantity of visit decreases from Q_0 to Q_1 and price increases from P_0 to P_1.

Paper Source
Subject.........
Date:.........Time:.........

(d) The virtual elimination of smoking on the market for hopital services.

When more people engage in smoking, then the demand for hopital services will increase. Therefore, its cost will increase. If people eliminate smoking, they will demand less hospital services, the price of per bed used reduces. Here, suppose that hospital service equals to the number of beds or cabin used.



Fig(4): Market for hospital service

As people reduce smoking, the demand for hospital services also decrease. So the demand curve for hospital services shifts to the left from D_H to D_H'. Initially, at D_H demand curve, the quantity of bed used was Q_1 and the price per bed used was P_1. With the shift of the demand curve, the equilibrium point shifts to E' where the quantity of bed used decreases to Q_1' and the price per bed used also decrease to P_1'.

Paper Source
Subject..................
Date:..................Time:..................

(e) Increased graduation of new doctors on the market for physician services.

Increased graduation of new doctors on the market for physician services increases the supply of physician services. So, this increases the quantity of visit and decreases the price of visit.

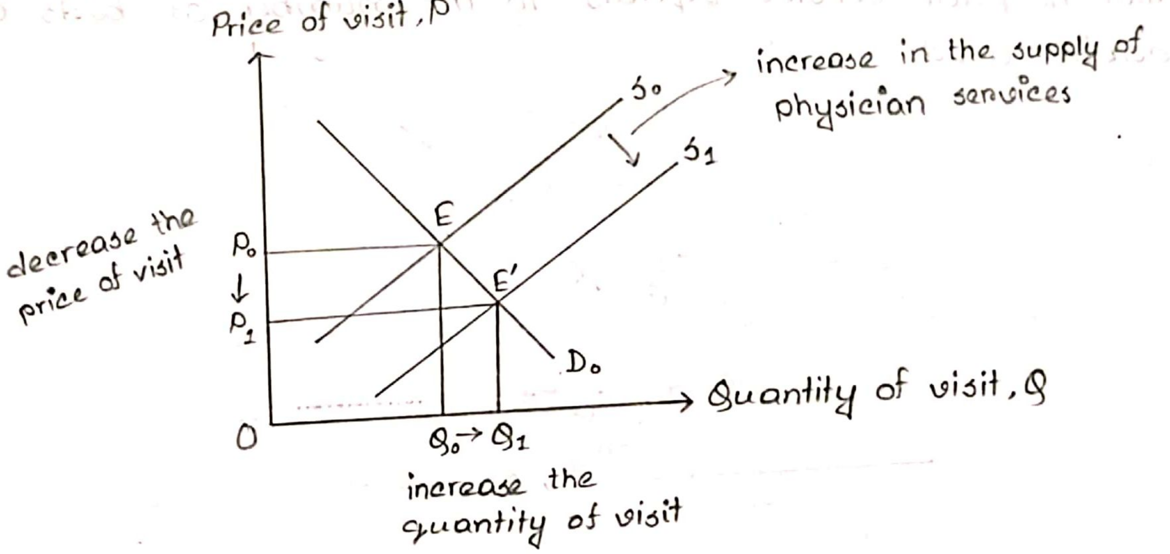
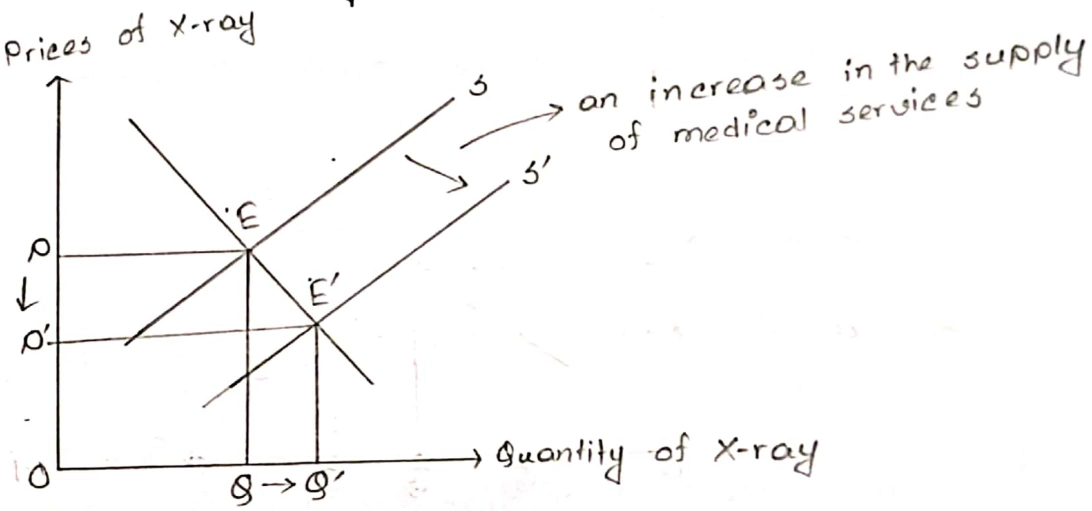


Fig:(5) Market for physician service

In the figure, when the graduation of new doctor increases, the supply of physician services increases. The supply curve for physician services shifts right from S_0 to S_1. As a result, the quantity of doctors visits increase from Q_0 to Q_1 and this leads to decrease in the price of visit from P_0 to P_1.

(f) A technological improvement that reduces the cost of producing X-rays on the market for physician clinic services.

A technological improvement that reduces the cost of producing X-rays on the market for physician clinic services leads to an increase in the supply of medical services.

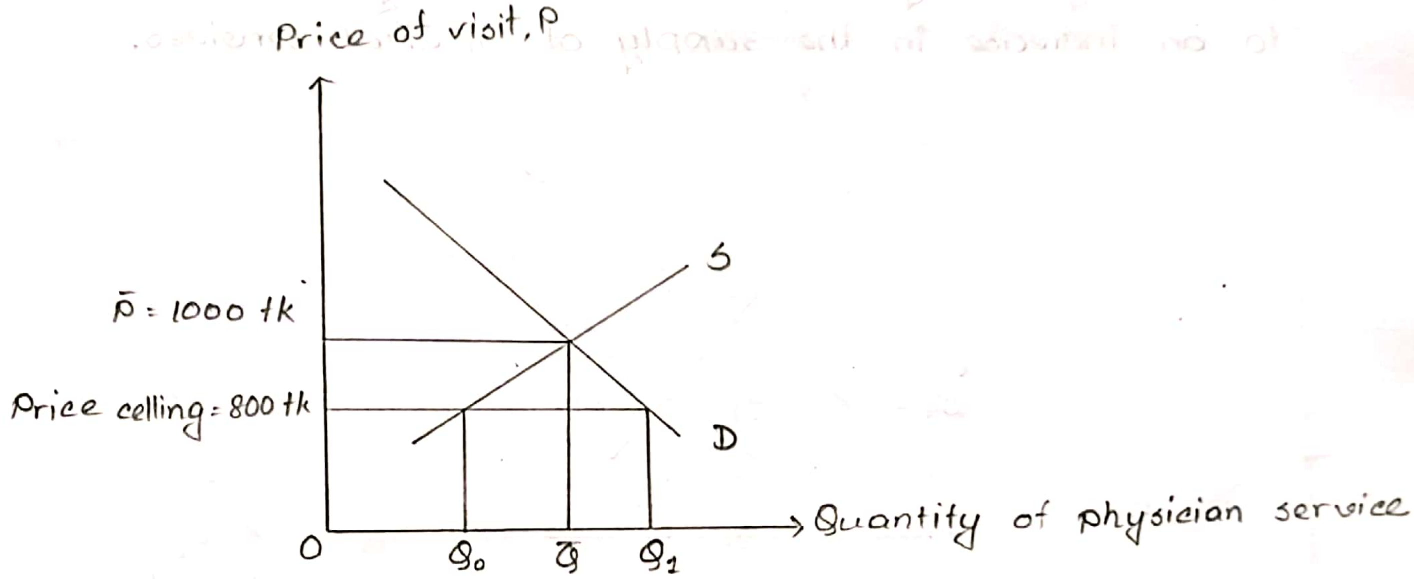


Fig(6): Medical service market

As the supply increases and the supply curve shifts from S to S', the price of X-ray decreases from P to P' and the quantity of X-ray increases from Q to Q'.

The price ceiling placed on physician fees in the market for physician service.

Without price ceiling the quantity of visit was $\bar{Q}$ at the 1000 tk price of per visit. Suppose, govt. placed a price ceiling on physician fees of 800 tk. This creates an excess demand for physician service, Q_1. But the supply of the physician services decreases to Q_0. As the demand for physician services increases and supply decreases, this decrease the quantity physician service.



Paper Source
Subject.................................
Date:.................................Time:.................

Define health production function. Describe the production function of health and marginal production function of health care. Would the production function of health ~~en~~ eventually bend downward?

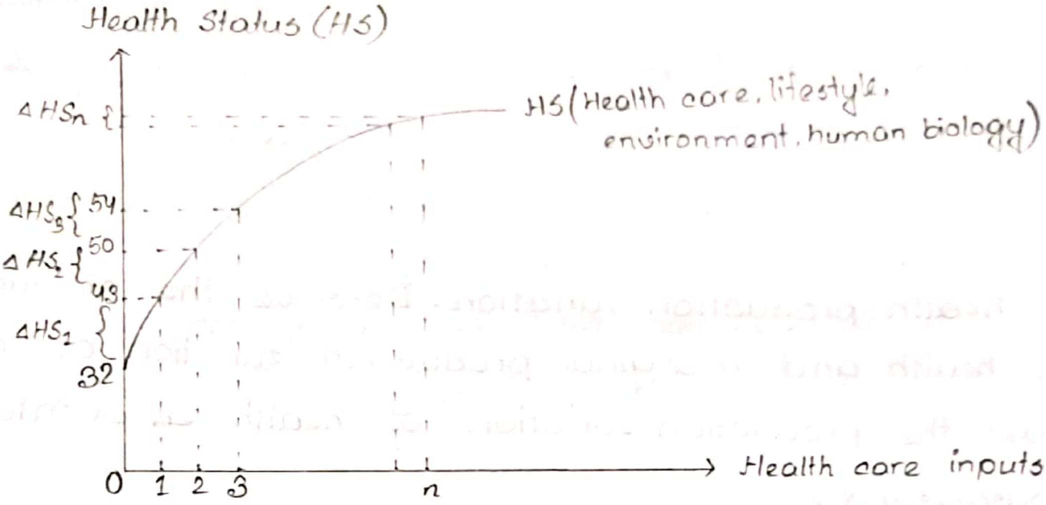
Health Production Fuction:

Health production function refers to the functional relationship between health care inputs and health status.

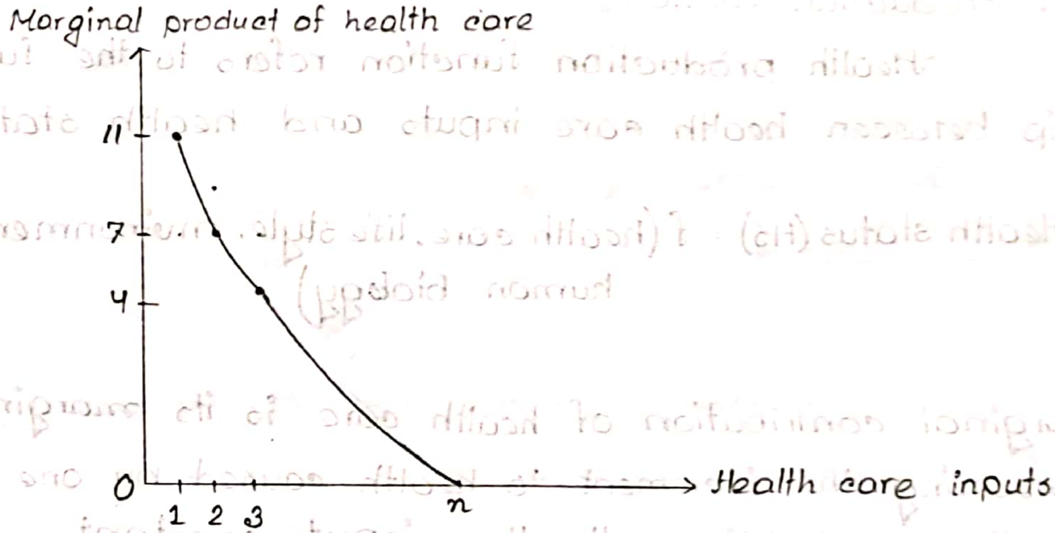
Health status (HS) = f (health care, life style, environment, human biology)

The marginal contribution of health care is its marginal product, meaning the increment to health caused by one extra unit of health care, holding all other inputs constant.

The distinction between total and marginal contribution of health care is illustrated in the following figures. Figure A exhibits a theoretical health status production function for the population. Health status is shown here to be an increasing function of health care. Improvement in any of health care inputs or factors will shift the curve upward.



A: Production function of Health



B: Marginal product of health care

Figure: Production of health

A production function describes the relationship of flows on inputs and output over a specific time period. Here, we assumed all health care inputs into one scale called Health care. In reality health care consists of many different health care inputs. Conceptually, the heath care measure, HC may be thought of as an aggregate of all these types of health care, the aggregation being based on dollar values.

Increasing health care from zero to one unit in figure (A) improves health status by ΔHS_1, the first unit's marginal product. Numerically, the first unit of health care has increased the health status increased from 32 to 43, $\Delta HS_1 = 11$, health status units. The next unit of medical care delivers a marginal product of $\Delta HS_2 = 7$ and so on.

These marginal products are diminishing in size, illustrating the Law of diminishing marginal returns. If society employs a total of n units of health care, then the total contribution of health care is the sum of the marginal products of each of the n units. Thus, the total contribution is shown as AB, may be substantial. However the marginal product of the nth unit of medical care is ΔHS_n and it is small. In fact, we are nearly on the 'flat of the curve." Marginal product is graphed on figure (B).

Downward bending of health production function:

We have drawn in figure the production function as a rising curve that flattens out at higher levels of health care but never bends downward. But the health production function eventually bends downward, that is, it is possible to get too much health care and so the health of the population is harmed. This is a logical probability under at least two scenarios —

(i) Iatrogenic (provider caused) disease.

This is an inevitable by product of many medical interventions. For example, each surgery has its risks.

(ii) Combinations of durgs may have ~~forseen~~ unforeseen and adverse interactions.

Many medical scientists such as Cochrane (1972) have pressed the case that much medical case as often practiced has only weak scientific basis, making iatrogenesis a real probability. Illich argued that medicalization would lead to less personal effort to preserve health and less personal determination to preserve; the ~~health~~ result becomes a decline in health of the population and thus a negative marginal product for medical care.

Paper Source
Subject.......................................
Date:............................Time:....................

Ques: Explain two different theories of schooling.

Answer:

(i) Micheal Grossman's Theory:

Grossman's theory of demand entails a central role for education. Grossman ~~en~~ contends that better ~~education~~ educated person tends to be economically more efficient producers of health status. Better educated people understand the technology or have the knowledge-how needed to stay healthy. They also know better how to use medical and other market inputs and their own time to produce health. Under this view, the marginal transfer of resources makes good sense.

Much of the available research concentrates on formal schooling. Other health economists have put ~~forced~~ forward hypothesis under which schooling and health status are correlated only because they are ~~rie~~both related to one or more other factors.

(ii) Victor Fuchs' Theory:

Victor Fuch has suggested a theory that involves certain microeconomic aspects of time discounting, that is comparing the future of the presence. Decision makers must discount future benefits and cost of a social investment to make them comparable in terms of present values. People

Paper Source
Subject...
Date:.......................................Time:......

who seek out additional education tend to be those with lower discount rates. A decision maker with a higher discount rate will tend to prefer projects with immediate payoffs versus long-term projects. People with a lower discount rate tend to be those who value the long-term gains more. Individuals with relatively low discount rates will be more likely to invest in education and in the health as well.

Paper Source
Subject.......
Date:.......Time:.......

Ques: According to Grossman model, how health care demand is different from traditional approach of demand?

Explain the demand for health and health care according to Grossman and comparison between traditional demand and health care demand.

Ans: Grossman used the theory of human capital to explain the demand for health and health care. According to human capital theory, individuals invest in themselves through education, training and health to increase their earnings. Health investment leads to a myraid of decisions regarding work, health care, exercise and the use of consumption goods and bads (such as unhealthy food, cigaretts, alcohol, or addictive drugs).

Grossman shows the way in which many important aspects of health demand differ from the traditional approach of demand —

(i) It is not medical care as such that the consumers want, but rather health. In seeking health, they demand medical care inputs to produce it.

(ii) The consumer doesnot merely purchase health passively from the market. Instead, the consumer produces health, combining time devoted to health improving efforts including diet and exercise with purchased medical inputs.

(iii) Health lasts for more than one period. It does not depriciate instantly and it can be analyzed like a capital good.

(iv) Perhaps, most importantly, health can be treated both as a consumption good and an investment good. As a consumption good, health is desired because it makes peopl feel and look better. As an investment good, health is desired because it increases the number of health days available to work and earn income.

These are the reasons of ~~two~~ how health care dem is different from traditional approach of demand.

Paper Source
Subject.
Date: Time:

Ques: Explain home good function and health investment function in the light of Grossman model.

Ans: An increment to capital stock, such as health, is called an investment. Consider a consumer Edward who buys market inputs (e.g. medical care, food, clothing) and combines them with his own time to produce the stock of health capital that produces services that increases his utility. Edward uses market inputs and personal time both to invest in his stock of health and to produce other things that he likes. During each period, Edward produces an investment in health, I. Thus,

$$\text{Health Investment, } I = I(M, T_H)$$

Here, M = Market health inputs (providers' service, drugs)

T_H = Time spent improving Health

This function indicates that, increased amount of M and T_H increase I.

There are also other items include virtually all other things that Edward does. They include time spent watching television, reading, playing with and teaching his children, preparing meals, baking bread or watching the sun set, a composite of other things people do with leisure time. We shall call this composite home good, B.

$$\text{Home good function, } B = B(X, T_B)$$

Here, X = market purchased goods

T_B = Time for producing home goods

Paper Source
Subject.......
Date:.......Time:.......

Home good (B) is produced with time, T_B and market purchased goods, X. This function indicates that increased amounts of X and T_B increase B.

Thus, Edward is using money to buy health care inputs, M and ~~good~~ home good inputs, X. He is using leisure time either for health care (T_H) or for producing the home good (T_B).

In this model, Edward's ultimate resource is his own time. Assume that, Edward has 365 days available in the year. To buy market goods such as medical care, M or other goods X, he must trade some of his time for income; that is he must work at a job. Call his time devoted to work T_W, we realize that some of his time during each year will be taken over by ill health or T_L. Thus we account for his total time in the following manner.

Total time = T = 365 days = T_H (Improving health) + T_B (Producing home goods) + T_L (Lost to illness) + T_W (working)

Paper Source
Subject.......
Date:....... Time:.......

Ques: How would you explain the potential use of an individual's time by labor-leisure trade-off? Show the equilibrium of an individual's time labor-leisure line and indifference curve.

Answer:

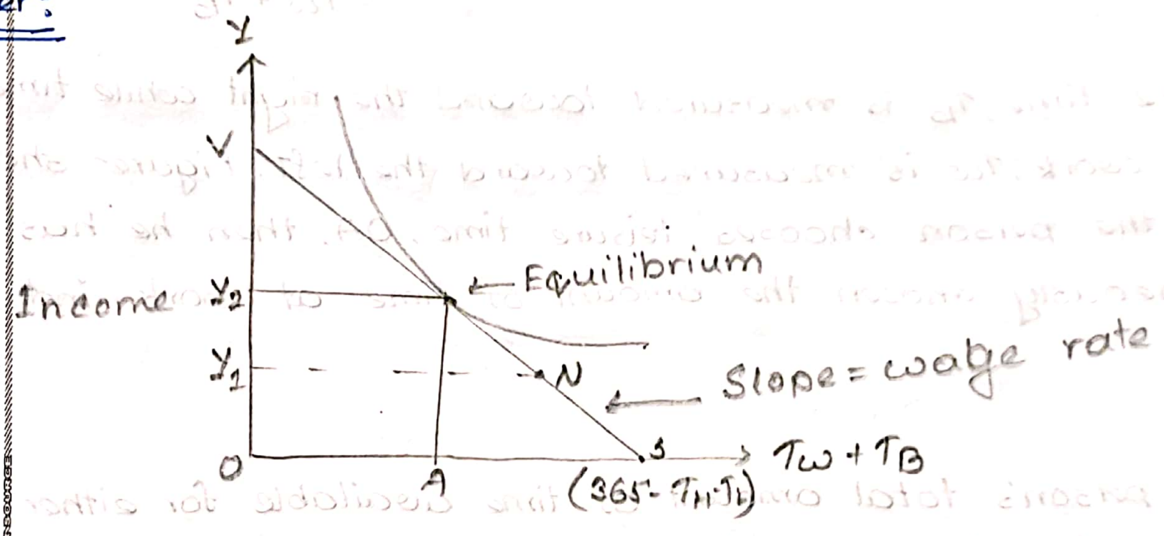


Fig: Labor-Leisure Trade-off

The potential uses of a person's time are illustrated by the labor-leisure trade-off. In the figure, The X-axis represents one's work and leisure time. Suppose that a person considers him time spent creating health investment to be "health-imporvement time" and that he calls T_B his leisure. Assume that the number of days lost to ill health and the number of day spent on health enhancing activity are fixed variable.

T_{Lo} = Time Lost

T_{Ho} = Time spent on health activities.

Paper Source
Subject.............................
Date:......................Time:..............

The maximum amount of time that he has available to use either for work, T_w or leisure, T_B is,

$$\text{Time available for work or leisure} = 365 - T_{H0} - T_{L0}$$
$$= T_w + T_B$$

Leisure time, T_B is measured toward the right while time spent at work, T_w is measured toward the left. Figure shows that if the person chooses leisure time, OA, then he has simultaneously chosen the amount of time at work indicated by AS.

The person's total amount of time available for either work or leisure is given by point S. If he chooses point S for the period, he would be choosing to spend all this available time in leisure. The y-axis represents income, obtained through work. This income will then purchase either market health goods or other market goods. Thus, if he chooses point S, he will not be able to purchase market goods because he has no wage income.

If the person gives up one day of leisure by working to point N, he will generate income equal to OY, which represents his daily wage. In economic terms, this

Paper Source
Subject
Date: Time:

quantity represents income divided by days worked, that is, the daily wage.

$$\text{The daily wage} = \frac{\text{Income}}{\text{Days worked}}$$

The slope of the line VS depeiting the labor-leisure trade-off reflects the wage rate. In equilibrium, person's trade-off of leisure and income is the same as the market trade-off, which is the wage rate. Here, he takes of OA amount of leisure and trades AS amount of leisure for income OY_2.

Ques: How is it possible to earn more money and more leisure time together? Explain, in the light of Grossman model.

Explain the impact of an increase in ~~act~~ activities to improve health on income and leisure.

Answer:

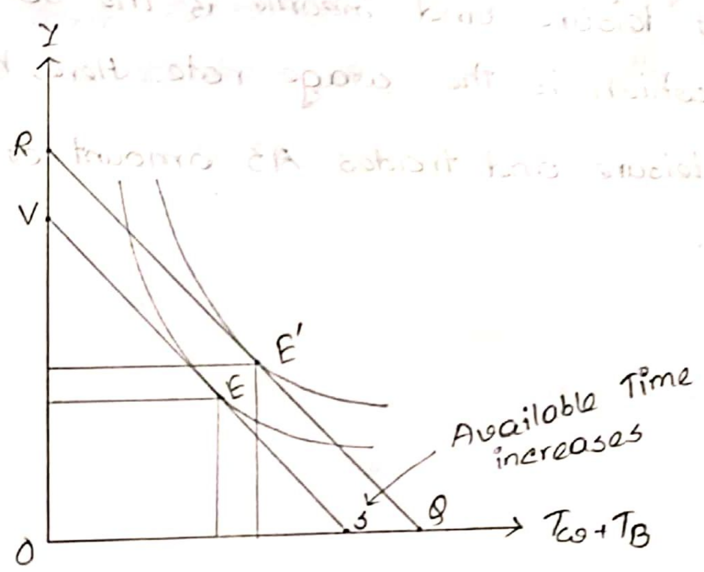


Figure: Increase amount of healthy time due to investment

An increase in activities to inprove health, increased the time spent investing in health status. Time spent of health producing activities T_H is increased from T_{H_0} to T_{H_1} correspondingly. Suppose that the number of days to lost to ill health has been reduced from T_{L_0} to T_{L_1}. Time

Paper Source
Subject:
Date: Time:

spent producing health reduces time available for other activities. On the other hand, time spent on health investment increases health stock and in turns reduces the time lost to illness.

If the net effect of $T_H + T_L$ is a gain in available time, then this illustrates the pure investment aspect of health demand. The health investments "pay off" in terms that not only add to ~~pro~~ potential ~~income~~ leisure and also increase the potential income, shifting the income-leisure line outward from VS to RQ. The expenditure of time for health producing activities may later improve ~~ab~~ available hours ~~of~~ of productive activity.

As a result of investment, utility increases moving from point E to E'. Not only does investment in health lead to feel better but also leads to more future income and may lead to more leisure as well.

The improvement in health status also might increase productivity of work, perhaps resulting in a higher wage.

Paper Source
Subject..........
Date:.......... Time:..........

Ques: Explain in the light of grossman model, the impact of age and education and wage on the optimum stock of health.

Answer:

Age: Age is the key factor for changing health condition. Health stock may depreciate faster during some periods of life and more slowly than others. Eventually, as he ages, the depreciation rate will likely increase.

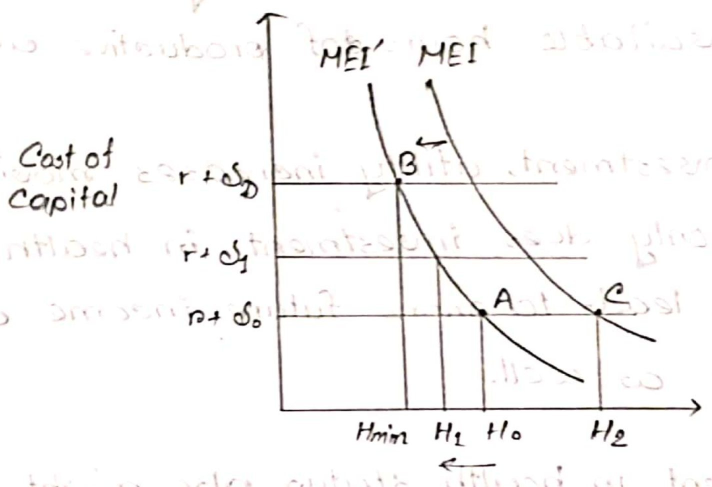


Fig. 1: Effect of age in person's health stock.

From the fig-1, we see by hypothesis, the depreciation rate increases from δ_0 to δ_1 and ultimately to δ_D. These assumptions imply that the optimal health stock decrease with age. The optimal health at younger age, H_0, is greater than H_1, the optimal Health at older age. Higher depreciation rates increase the cost of holding capital stock. In Old age, health depreciation rates are extremely high, δ_D and optimal health stock fall to H_{min}, at point B. This suggests that as expected length of life decreases, the MEI curve shifts to the left, because the returns from an investment will last for a shorter period of time.

Wage rate:

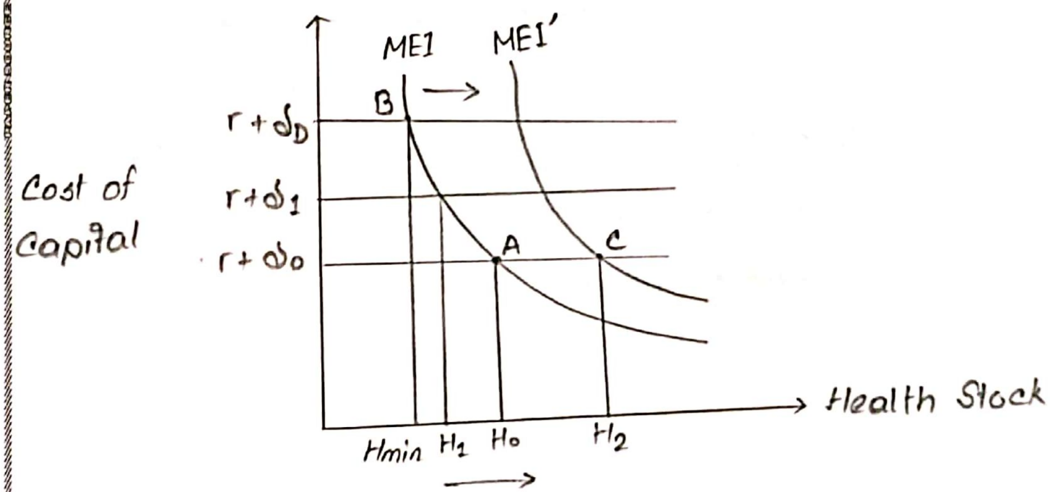


Fig 2: Effect of increasing wage rate.

Fig 2 illustrates the effect of a change in the wage rate. Increased wage rates increase the returns obtained from healthy days. Thus, higher wages imply a higher MEI curve or MEI'.

Paper Source
Subject.........
Date:......... Time:.........

Ques: How optimum health stock is determined? Explain with diagram.

Answer: The MEI can be described in terms of the X-ray machine. Suppose, a X-ray machine price is \$100000 and annual income from it is \$20000, that the rate of return, $i = \frac{20000}{100000} = 20\%$. The management would buy this machine if rate covers its opportunity costs and depreciation cost.

If management considered owning two machines, then the second machine would have a lower rate of return than the first. Because first machine would assign it to the highest priority uses. those with the highest rate of return. And second machine would assign it to the lesser priority uses (might be idle occasionally). The clinic will purchase the second machine only if its rate of return is still higher than the interest plus depreciation.

The decreasing MEI:

In the figure, the marginal efficiency of investment curve, MEI, degcribes the pattern of rates of return. declining as the amount of investment increases. The cost of capital (that is the interest r, plus the depreciation rate, δ_0) is shown as the vertical line labeled $r+\delta_0$. The optimum amount of capital demanded is thus H_0, which represents the amount of capital at which the marginal effeciency of investment just equals the cost of capital. This equilibrium occurs at point A.

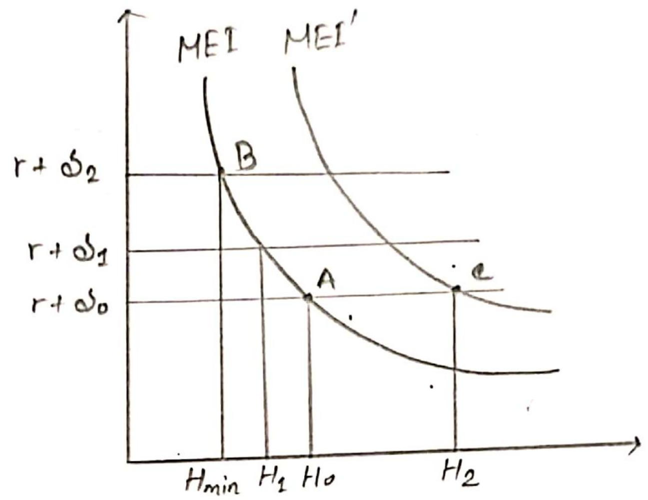


Fig: Optimal Health Stock.

The MEI curve for investment in health also would ~~de~~ be downward sloping. This occurs because the production function for healthy days exhibits diminishing marginal returns. The cost of capital for health would similarly reflect the interest rate plus the rate of depreciation in health. Understand that a person's health, like any capital good, also will depreciate over time. ~~As we~~ Thus, the optimal demand for health is likewise given at the intersection of MEI curve and the cost of capital curve $(r+\delta_0)$.